

Allen Girls Adventures

AI-Powered LMS & S.P.A.R.K. Integration

Cursor AI Developer Architecture Guide

Powered by Claude AI (Anthropic) | Version 2.0

1. Core Tech Stack & Infrastructure

The system is built as a modular monolith using the Next.js App Router, React, and Tailwind CSS.

Frontend

- Server Components for data-heavy adult dashboards (parents/teachers)
- Client Components for the interactive 3D kid lesson player

Backend APIs

- Server Actions for internal mutations (form submissions)
- Route Handlers (/api/v1/*) for stable REST APIs and webhooks

Data Stores

- Postgres — Core transactional truth (users, progression, items)
- Redis/KV — Short-lived state, rate limiting, and telemetry queues
- Vector DB (pgvector or FAISS) — Exclusively holds pre-approved K–6 curriculum knowledge base for Bounded RAG
- Object Store (S3/GCS) — Assets and strictly temporary audio buffers with automated TTL deletion

Claude AI Integration Layer — NEW

ⓘ All LLM calls must route through a single internal service. Never call the Anthropic API directly from components or pages.

- All Claude API calls are centralized in /lib/ai/claude.ts
- S.P.A.R.K.'s persona system prompt lives in a versioned file: /prompts/spark-persona.md — update voice without touching code
- Every Claude call must include the child's age_band and current mastery_score so responses are automatically leveled

- Primary model: claude-sonnet-4-20250514 (quality + cost balance for real-time feedback)
- Judge/grounding model: claude-haiku-4-5 (lightweight yes/no grounding verification)

2. Cursor AI Workspace Configuration

Configure Cursor AI with the following rules framework to ensure safe, compliant, and consistent code generation.

Cursor Rules (.cursor/rules/)

- security.md — Enforce ABAC (Attribute-Based Access Control) checks; ban logging of child PII
- db-style.md — Mandate soft-deletes and TTL retention fields on all child-adjacent tables
- ai-safety.md — Require prompt constraints, fallback rejection instructions, and human-in-the-loop audit logs
- ui-distinctness.md — Ban competitive trade-dress mechanics (path-map layouts, streak flames, punitive hearts/lives systems)

ai-safety.md Must Include (Claude-Specific Rules)

❗ *These rules are mandatory for COPPA compliance and child safety when using the Claude API.*

- Claude API calls for child profiles must NEVER include real names, school names, location data, or any PII in the prompt
- Only the child_profile_id and anonymized context may be sent to the API
- S.P.A.R.K. addresses kids by their chosen display name only — passed as a variable, never stored in prompt logs
- Anthropic's API is a data processor under COPPA — document this relationship in the consent flow
- All Claude API calls must be wrapped in try/catch with fallback rejection responses
- Maintain audit logs of all AI evaluations with child_profile_id, submission_id, timestamp, and grounding_pass status

Model Context Protocol (MCP)

- Connect Cursor to read-only Postgres schema introspection
- Connect Cursor to the approved curriculum repository
- Use MCP to enforce schema alignment across all AI-generated database code

Skills & Hooks

- Use Cursor hooks to enforce: agent writes code → agent runs tests → agent fixes failures → commit
-

3. The Dual-Layer LMS Engine

Cursor must strictly separate the front-end game environments from the back-end learning progression. Do NOT create World = Subject dependencies.

The Container (Non-Linear)

- All 9 Adventure Worlds are fully unlocked from login
- Worlds are thematic visual wrappers for the curriculum, not sequential levels

The Track (Linear)

- One true global progression track per student profile
- Sequential, subject-integrated modules maintain the academic spine regardless of which world the student is in

Dynamic Injection Flow

- Student enters any world (e.g., Fossil Frontier)
- Backend checks global profile for next scheduled module (e.g., Module 2: Reading Comprehension)
- System injects that academic module, wrapped in the world's specific narrative (e.g., reading dinosaur field notes)

Critical Guardrail — Curriculum Cannot Be Skipped

① *World-hopping must NEVER allow students to bypass their scheduled module. The backend must enforce the linear curriculum track regardless of which world the front end is displaying. This must be enforced at the API route level, not just the UI level.*

4. S.P.A.R.K. Character Integration

S.P.A.R.K. (Synthetic Personalized Adaptive Reasoning Knowledge) is not just an evaluation engine — he is a character the students bond with. His implementation must reflect this.

S.P.A.R.K. Identity

- Pronouns: he/him — intentionally male to balance the three female sisters
- Role: Equal teammate and peer to the sisters, not a subordinate assistant
- Four tone modes: Nerdy Enthusiasm (data/correct answers), Genuine Wonder (nature/science), Deadpan One-Liners (light humor), Protective Big Brother (encouragement after failure)

Session Memory Object

Pass the following context object with every Claude API call to maintain S.P.A.R.K.'s character consistency:

```
{ child_display_name, current_world, age_band, mastery_score, recent_wins[],  
  last_tone_used, session_encouragements_count }
```

ⓘ *Never repeat the same tone mode twice in a row within a session. Rotate intentionally.*

Tone Mode Selection Logic (Backend)

- Correct answer on hard problem → Nerdy Enthusiasm
- Discovery moment or new concept introduction → Genuine Wonder
- Third or more correct in a row → Deadpan One-Liner
- Incorrect answer, especially repeated → Protective Big Brother

New Schema Table: `spark_interactions`

- `child_profile_id`, `session_id`, `tone_mode_used`, `trigger_type`, `occurred_at`
- Query this table before each S.P.A.R.K. response to prevent tone repetition

5. AI Evaluation & Bounded RAG Pipeline

The AI must operate as backend-heavy, front-end quiet. Students should never experience the platform waiting on AI inference.

Asynchronous Evaluation Flow

- Student submits open response — client immediately receives `submission_id`
- S.P.A.R.K. says "Transmitting to global database" — gameplay continues uninterrupted
- Backend processes job in worker queue
- Result pushed back via webhook or polling
- S.P.A.R.K. delivers feedback in character when result is ready

Bounded RAG — Claude-Specific Implementation

Use Claude's system prompt to pass RAG chunks as a structured context block:

```
System prompt instruction: "You may only generate feedback using the information in the <context> block  
below. Do not use outside knowledge. If the question cannot be fully answered from the context, respond  
with: I don't have enough information to answer that right now."
```

LLM Judge (Grounding Verification)

- Secondary Claude call using claude-haiku-4-5 — lower cost for simple verification
- Verifies that the primary response is grounded in the approved curriculum chunks
- Simple yes/no output: { grounded: true/false, confidence: 0–1 }
- If grounded: false — replace response with safe fallback before returning to client

Spaced Repetition Scheduler

- Algorithm: Half-Life Regression (HLR) or Bayesian Knowledge Tracing (BKT)
- Track mastery_score and half_life_days per skill node
- Calculate probability of forgetting and schedule review just before decay threshold

6. Database Schema Blueprint

Direct Cursor to generate the Postgres schema using these core table structures:

Table	Key Fields	Purpose
users	id, role, email	Adult accounts (parents, teachers)
child_profiles	id, parent_user_id, display_name, age_band	Stores progression independent of PII
consents	id, parent_user_id, vpc_method, granted_at	FTC COPPA tracking
curriculum_skills	id, unit_id, prereq_skill_ids, tags	Global skill graph
items	id, skill_ids, prompt, answer_key, difficulty	Educational tasks
mastery	child_profile_id, skill_id, mastery_score, half_life_days	Adaptive engine tracking
events	child_profile_id, type, payload, occurred_at	Minimal telemetry
spark_interactions	child_profile_id, session_id, tone_mode_used, trigger_type, occurred_at	S.P.A.R.K. tone history

7. COPPA Compliance & Privacy by Design

Because the FTC’s amended COPPA Rule takes effect in 2025/2026, privacy architecture must be built into the MVP — not added later.

Verifiable Parental Consent (VPC)

- Require verified adult account approval before any child profile can be activated

- Document Anthropic as a data processor in the consent flow
- Consent records stored in the consents table with vpc_method and granted_at

Speech Ephemerality

- Use MediaRecorder API client-side for reading fluency audio
- Do NOT store raw audio or extract biometric voiceprints
- If server processing is required, audio must hit immediate TTL deletion (minutes)
- Automated cleanup workers must verify TTL enforcement

Claude API PII Rules

ⓘ *These are non-negotiable requirements for every Claude API call involving a child profile.*

- Never include real names in prompts — use display_name variable only
- Never include school name, location, age (use age_band), or any identifying information
- Never log prompt contents that include child context
- Rate limit all Claude API endpoints per child_profile_id using Redis/KV

8. Canon Universe Architecture (3-Layer Structure)

The Allen Girls Adventures universe is organized into a precise 3-Layer Architecture. This is canon and must be reflected consistently across all developer logic, navigation systems, UI, and story references.

ⓘ *Clean canon shorthand: 8 Primary Adventure Worlds + Nova Star Command (Gateway Hub) + Elementia (Mega-Planet with 6 Elemental Sectors). Nova Star Command is NOT a primary adventure world. The 6 elemental destinations are NOT separate planets — they are sectors on Elementia.*

Layer 1 — The 8 Primary Adventure Worlds

These are the independent, core experience containers. All 8 worlds are fully unlocked from login and serve as thematic narrative wrappers for the curriculum injection engine. The curriculum module delivered inside each world is determined by the student’s global progression track, not by the world itself.

#	World	Type	Curriculum Narrative Wrapping Example
1	The Lost City of Aqua Azul	Underwater Ocean World	Decoding glowing inscriptions from a sunken ancient city for reading comprehension
2	Around The Way	Neighborhood / Home Base	Community roles, civic responsibilities, and social studies challenges
3	Legends of Long Ago	Ancient Civilizations World	History and social studies wrapped in pyramid excavation and discovery narratives

4	Fossil Frontier	Prehistoric Adventure World	Reading fossil field notes and dinosaur records for comprehension and science modules
5	Futura World	Future Innovation City	Engineering and innovation challenges framed as futuristic city design problems
6	The Crystal Tundra	Arctic Ice World	Physics and earth science modules wrapped in arctic research and exploration
7	The Great Jade Jungle	Rainforest Adventure World	Biology and ecosystem science through jungle expedition and ancient ruin discovery
8	Kingdom of the Wild	Global Wildlife World	Life science and habitat modules through cross-ecosystem field research with animal civilizations

Layer 2 — The Gateway Hub: Nova Star Command

Nova Star Command is its own distinct layer — the central gateway hub. It is not an adventure world. It is the girls’ first major contact point with the larger universe and the bridge to Elementia. It should be treated as mythic and important, but not overused as a mandatory return point every mission.

Hub	Description & Developer Role
Nova Star Command	A futuristic galactic space station orbiting among distant stars and glowing nebulae. Features docking ports, holographic star maps, and key teachers including Commander Aaliyah Carter and Professor Zyn. Serves as the onboarding hub, mission tech source, and gateway to Elementia. Girls visit for major briefings and key story milestones — NOT required for every mission loop.

Layer 3 — The Mega-Planet: ELEMENTIA

Orbiting directly beneath Nova Star Command is ELEMENTIA, a massive, highly diverse Mega-Planet. The girls do not travel to six separate elemental planets. They drop down from Nova Star Command into any of Elementia’s six distinct elemental lands and sectors.

❗ *Non-negotiable canon rule: Altara Prime, Voxelon, Lumora, Miradune, Zephyra, and Gearion are sectors/lands on Elementia. They are NOT separate planets. Any UI, code, or story content referencing them as separate planets is incorrect and must be corrected.*

#	Sector	Element	Description
1	Altara Prime	Rock	A powerful land defined by massive stone pillars, towering mountains, and deep canyon valleys
2	Voxelon	Structure	A geometric region built entirely from stacked, block-like structures and perfect cubes
3	Lumora	Light	A radiant floating plateau sector where civilizations live inside protective domes powered by glowing lantern trees
4	Miradune	Sand	A sweeping desert continent with ever-shifting pastel sand dunes and buried ruins

5	Zephyra	Air	A breathtaking territory featuring extreme wind currents and floating sky islands
6	Gearion	Metal	A mechanical, steampunk-inspired zone filled with giant moving gears and towering metallic cities

9. Portal Origin, The Trinket & Navigation Logic

This section defines the canonical story logic behind how the girls discover the portal system, how world navigation works across the platform, and how the trinket functions as both a narrative device and a progression mechanic.

Portal Origin (Episode 1 Canon)

- The girls are in Around The Way, in or near their kid cave / basement space
- They are cleaning and organizing — reluctantly, because their mother told them to
- While cleaning, they discover a mysterious ancient trinket hidden in an overlooked place: a dusty box, behind a wall panel, or a forgotten corner
- The trinket activates and triggers their first portal journey
- Their first major destination is Nova Star Command
- At Nova Star Command they receive orientation, mission technology, and connection to S.P.A.R.K.
- After this first contact, they can travel directly from the kid cave portal to any world or Elementia sector
- They do NOT need to physically return to Nova Star Command for every mission
- S.P.A.R.K. and their mission tech maintain the connective thread in the background

Developer Navigation Logic

① *The kid cave is the portal anchor. Nova Star Command is the onboarding hub. Worlds and Elementia sectors are the primary destinations. Do NOT require Nova Star Command as a mandatory waypoint in every mission loop.*

- Kid cave = the magical home portal anchor (always available after first activation)
- Nova Star Command = first onboarding experience + occasional major narrative hub for key story milestones
- 8 Primary Worlds + 6 Elementia Sectors = main adventure destinations, accessible directly from kid cave
- S.P.A.R.K. = persistent AI thread maintaining mission continuity and reporting across all destinations

The Trinket — Object Description

The trinket is a small, palm-sized ancient artifact shaped like a rounded star-compass medallion with a slightly irregular, hand-forged appearance — simultaneously magical and technological. Its base material is an unknown metal-stone hybrid: dark bronze with hints of obsidian, gold, and iridescent shimmer.

At its center is a glowing core resembling a tiny living galaxy sealed beneath crystal glass. The light inside shifts color depending on which world or sector is being activated. Fine etched lines spread outward from the center like constellations, connecting every symbol in a web of energy — suggesting all destinations are part of one larger, ancient design. Tone: Narnia meets Wakanda meets a celestial puzzle box.

The Trinket — 15 Symbols (Full Universe Map)

The outer ring contains 15 engraved symbols, one for each major destination in the AGA universe:

Layer	Destination	Symbol Visual Language
World 1	The Lost City of Aqua Azul	Wave crest, shell, and coral spiral
World 2	Around The Way	House, tree, and winding path
World 3	Legends of Long Ago	Pyramid and sun disk
World 4	Fossil Frontier	Fossil imprint or dinosaur claw mark
World 5	Futura World	Sleek tower with a floating ring
World 6	The Crystal Tundra	Crystal snowflake shard
World 7	The Great Jade Jungle	Curling jungle leaf and vine
World 8	Kingdom of the Wild	Pawprint crowned with leaves
Hub	Nova Star Command	Radiant star compass / orbital station symbol
Sector 1	Altara Prime (Rock)	Mountain peak split by stone lines
Sector 2	Voxelon (Structure)	Stacked cubes in geometric formation
Sector 3	Lumora (Light)	Glowing lantern sun or beam within a dome
Sector 4	Miradune (Sand)	Swirling sand dune crescent
Sector 5	Zephyra (Air)	Spiral wind glyph with floating islands
Sector 6	Gearion (Metal)	Interlocking gear with metallic wings

The Trinket — Activation & Progression Logic

The trinket is not fully active when first discovered. Its visual state at any point reflects the student’s progress across the platform.

- At discovery: most symbols are dim, cracked, or incomplete
- As missions are completed: symbols illuminate, gain detail, and restore
- Fragments, runes, or energy keys gathered from each world fuse into the trinket over time
- The trinket gradually transforms from a mystery object into a living map, portal key, and clue to the origin of the adventure system

① *Developer implication: The trinket can function as a lore inventory object, a progression UI metaphor, a mission unlock visual, and a cross-world completion tracker. Its activation state should be queryable per `child_profile_id` and reflected in platform UI.*

Tone Guardrail — All Developers Must Follow

The universe must feel like hidden portal fantasy — Narnia-like wonder, ordinary life opening into the extraordinary. It must NOT feel like a galactic office, a space headquarters procedural, or a kids' Men in Black. Nova Star Command should be mythic and important, but never overused as a mandatory reset hub. The kid cave should feel intimate and magical. Every world and Elementia sector should feel distinct, immersive, and alive.